

1)

Steps:

1. $h(n) = \# \text{ incorrect letters}$
- 2.

The steps cost measure operates by looking at how many incorrect letters remain in the current iteration of the word. Each action the agent takes in this measure costs 1 and the only available action is letter replacement at an arbitrary but particular index. Thus, navigating to the goal state requires at least an equivalent cost to the number of incorrect letters. This heuristic, however, is relaxed to not consider whether a word is valid as the initial problem outlines. Thus, the unavoidable remaining cost which this heuristic lower-bounds is the additional cost picked up by traversing an indirect path of valid words. This entails any index needing to be altered more than once to arrive at the goal. An example of this can be seen when going from 'ask to 'why', where the third index transmutes along the path of k, s, a, then y.

3.

The actual optimal path to the goal state must either be a direct path as outlined by the heuristic or an indirect path, which must have a greater cost than the heuristic as all traversals cost 1. Therefore, the true optimal path of any state transition, from A to B , must be greater than or equal to the given heuristic. Thus, the heuristic cannot overestimate the true optimal remaining cost.

Scrabble:

1. $h(n) = \sum \text{value}(\text{goal}_i), n_i \neq \text{goal}_i$
- 2.

The scrabble cost measure utilizes the sum of the corrected letters' values from the current to goal state in determining cost. Each letter has been given a predetermined integer cost with the only available action being letter replacement at an arbitrary but particular index. Thus, navigating to the goal state requires at least an equivalent cost to this sum. This heuristic, however, is relaxed to not consider whether a word is valid as the initial problem outlines. Thus, the unavoidable remaining cost which this heuristic lower-bounds is the additional cost accrued traversing an indirect path of valid words. This entails any index needing to be altered more than once to arrive at the goal. An example of this can be seen when going from 'ask to 'why', where the third index transmutes along the path of k, s, a,

then y. While a direct change from k to y would cost 4, the addition of s and a, both cost 1, causes the actual cost to be 6.

3.

Like the steps cost measure, the actual optimal path to the goal state utilizing scrabble must either be a direct path as outlined by the heuristic or an indirect path. As all traversals have a cost greater than or equal to zero, any indirect path must have a greater cost than the heuristic. Therefore, the optimal path of any state transition, from A to B , must be greater than or equal to the given heuristic. Thus, the heuristic cannot overestimate the true optimal remaining cost.

Frequency:

1. $h(n) = n == \text{goal} \rightarrow 0, (1 + \text{goal frequency}) + (\# \text{ incorrect letters} - 1) * (1 + \text{minimum frequency})$

2.

The frequency cost measure collapses the frequency values down to their minimum. It then utilizes the minimum number of iterations required to reach the goal, though the number of incorrect letters, to acquire the lowest possible cost. It achieves this by using the known cost of arriving at the goal summed with the assumption that every in-between stage will be the least valued frequency. Since it is impossible for every iteration to be this minimum cost, the unavoidable remaining cost this heuristic lower-bounds is the difference from the minimum to the actual frequency of each stage.

3.

The heuristic of this cost measure assumes an identical frequency across all words, that being the minimum. Under this assumption, with the heuristic considering the most direct path, any viable path to the goal, direct or indirect, must be at least equivalent to the heuristic. Furthermore, there would be no feasible way that the actual frequencies, which would be greater than or equal to the minimum, could sum to a value less than this ideal quantification. Therefore, the heuristic cannot overestimate the true optimal remaining cost.

2)

Word Pair	Cost Measure	UCS Cost	UCS Nodes Expanded	A* Cost	A* Nodes Expanded	% Reduction
why→ask	Steps	5	773	5	16	97.9301423
why→ask	Scrabble	9	765	9	13	98.30065359

why→ask	Frequency	38.85540652	808	38.85540652	754	6.683168317
bear→duck	Steps	4	691	4	6	99.1316932
bear→duck	Scrabble	11	2981	11	6	99.79872526
bear→duck	Frequency	38.2402284	1084	38.2402284	714	34.13284133

1. UCS and A* always returned an identical optimal cost.
2. For both word pairs, (why, ask) and (bear, duck), the cost measure in which A* produced the largest percentage reduction in expanded nodes was scrabble.
3. For both word pairs, (why, ask) and (bear, duck), the cost measure where the heuristic was least informative was frequency.

4.

The differences between the informative natures of each heuristic involve their closeness with the actual remaining cost to the goal state. With proximity the A* algorithm is provided with highly viable information it can use to prune branches and deduce the optimal path. For both the steps and scrabble heuristics, they are highly accurate to the actual problem, utilizing the mismatched letters to produce their values. In the case of steps, it gives the minimum distance from the current and goal states while scrabble provides the minimum transitional value between them. Despite utilizing minimums, these heuristics employ the underlying features of the problem, such as path length and letter value lookup tables, which keep them tight to the original problem and actual remaining cost.

On the other hand, the frequency heuristic utilizes a multiplicative approach of two minimums, of the frequency values and distance to goal, scaling the difference between the theorized and actual remaining cost. Additionally, utilizing only the minimum of the frequencies completely disregards the rest of the domain's data preventing them from aiding the heuristic in staying true to the original problem. Furthermore, the decent range between the minimum and maximum frequency values widens this gap even more. These three issues compiled cause this heuristic to be too relaxed from the original problem, making it weak and uninformative for the A* algorithm.

3)

Steps:

1. Yes
- 2.

When operating in the steps cost measure, all actions cost 1 irrelevant of what action is taken. This means that any reversed action, if reversable, will have a symmetric cost to the original, both being 1. The rules of this problem state that all

states must be valid words in the given dictionary and each action is a one letter substitution. Thus, any traversal must be a valid word to another valid word through a single letter substitution, which is a logically reversible operation. Combining these two principles, when traversing a path between states A and B , the length of that path, the number of actions taken, is the cost for both A to B and B to A . Therefore, the steps cost measure offers both reversibility of action and symmetry of cost.

3. N/A

Scrabble:

1. No
- 2.

For the scrabble cost measure, a lookup table which associates a letter key to an integer value is utilized to determine cost justified via the substitution letter. As these values are not uniform, an action which transmutes state A to B through letter substitution of x to y can result in a different cost from the reverse action, if reversible. The rules of this problem state that all states must be valid words in the given dictionary and each action is a one letter substitution. Thus, any traversal must be a valid word to another valid word through a single letter substitution, which is a logically reversible operation. This means that while all actions are reversible, they are not guaranteed to be symmetric in cost. Therefore, when traversing a path between states A and B , the optimal cost, the sum of the substitution values, from A to B and B to A may not be equivalent.

3.

icky→poop:

Optimal Cost: 11

$$2(n, 1) + (s, 1) + 2(o, 1) + 2(p, 3) \rightarrow 11$$

Path: icky inky inks inns ions pons poos poop

poop→icky:

Optimal Cost: 16

$$(s, 1) + 2(n, 1) + (i, 1) + (k, 5) + (y, 4) + (c, 3) \rightarrow 16$$

Path: poop poos pons ions inns inks inky icky

Frequency:

1. No
- 2.

The frequency cost measure defines cost by the target word's predetermined frequency as outlined in the given dictionary. This means that the actions of state A to B and C to B will both have an equivalent cost. However, the reversal of either action will likely not have equivalent cost to each other or the original value. Thus, only when two paths share an identical set of 'to x ' actions, will the paths be guaranteed to have equivalent cost. Given the optimal traversal paths between states A and B , A to B and B to A will have a cost differential equivalent to the difference between the frequency cost of B and A . However, each action is reversible based on the rules of the problem which states that all states must be valid words in the given dictionary and each action is a one letter substitution. This provides a logically reversible operation as any traversal must be a valid word to another valid word through a single letter substitution. Therefore, while frequency cost measure permits reversibility of actions, it does not ensure symmetry of its cost nor the path's overall cost.

3.

ask→why:

Optimal Cost: 37.76

(ass, 9.70) + (ais, 13.89) + (his, 0.97) + (has, 0.66) + (was, 0.47) + (way, 1.94) + (why, 3.13) → 37.76

Path: ask ass ais his has was way why

why→ask:

Optimal Cost: 38.86

(way, 1.94) + (was, 0.47) + (has, 0.66) + (his, 0.97) + (ais, 13.89) + (ass, 9.70) + (ask, 4.23) → 38.86

Path: why way was has his ais ass ask

frequency(ask) – frequency(why) = 4.23 – 3.13 = 1.1 = 38.86 – 37.76 =
cost(why→ask) – cost(ask→why)

4)

1.

Model: Claude Sonnet 5

Prompt:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering

word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Proposed Heuristic: $h(n) = \begin{cases} 0, & n = \text{goal} \\ (h_0(n) - 1)(1 + m) + (1 + \text{rarity}(\text{goal})), & h_0(n) \geq 1 \end{cases}$

- $h_0(n) = \#\{i: n_i \neq \text{goal}_i\} - \# \text{ incorrect letters}$
- $m = \min_{w \in D_L} \text{rarity}(w) - \text{cheapest word of the right length anywhere in the dictionary}$

2.

$h(n) = \begin{cases} 0, & n = \text{goal} \\ (h_0(n) - 1)(1 + m) + (1 + \text{rarity}(\text{goal})), & h_0(n) \geq 1 \end{cases}$

- $h_0(n) = \#\{i: n_i \neq \text{goal}_i\} - \# \text{ incorrect letters}$
- $m = \min_{w \in D_L} \text{rarity}(w) - \text{cheapest word of the right length anywhere in the dictionary}$

3.

Let $L_0 = \#\{i: n_i \neq \text{goal}_i\}$: the minimum length/moves of an optimal path from A to B

Let L_a = the actual length/moves of an optimal path from A to B

$| L_0 \leq L_a$

Let $h(n) = \sum_{i=0}^{L_0-2} (1 + \text{rarity}(\min)) + (1 + \text{rarity}(\text{goal}))$

Let $h^*(n) = \sum_{i=0}^{L_a-2} (1 + \text{rarity}(w_i)) + (1 + \text{rarity}(\text{goal})), w = \text{a zero-indexed word state between } A \text{ and } B$

By the definition of minimum, $\text{rarity}(\min) \leq \text{rarity}(w_x)$

$$\because \text{rarity}(\min) \leq \text{rarity}(w_x) \ \& \ L_0 \leq L_a \rightarrow \sum_{i=0}^{L_0-2} (1 + \text{rarity}(\min)) + (1 + \text{rarity}(\text{goal}))$$

$$\leq \sum_{i=0}^{L_a-2} (1 + \text{rarity}(w_i)) + (1 + \text{rarity}(\text{goal}))$$

$$\therefore h(n) \leq h^*(n), \forall n \in \{A \rightarrow B\}$$

By the definition of heuristic admissibility, this heuristic is admissible

4. N/A

5.

The AI's verified proposed heuristic and the heuristic initially deduced for the frequency cost measure are identical. As such, the solution cost and number of expanded nodes will be equivalent for the problems outlined. As to whether the heuristic should be utilized, it should not. While the heuristic is admissible, therefore usable, it is not significantly informative as seen through the percent reduction values and responses of Question 2. Instead, another heuristic which increases the percent reduction, through tightening $h(n)$ to $h^*(n)$, should be determined and utilized instead.